ACCESS · Indigenous Skills 4 Success

Asphalt paving · Roadbuilding foundations

Industry, materials, people, road documents and evidence.

Classroom teaching edition · 29 October 2026 · Technical review and pilot pending

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Asphalt paving · Roadbuilding foundations

Learn the industry, read the work and explain a decision

Roadbuilding connects materials, people, places and decisions. This book starts with the industry and moves through road history, paving technology, team responsibilities, road documents and emerging approaches. You will read evidence, work through examples, make calculations, discuss choices and record what still needs checking. The classroom cases give you the information needed to practise without using private workplace records.

This is the 29 October 2026 classroom teaching edition. Qualified technical review and classroom pilot are pending. Classroom completion records learning evidence; equipment training, workplace orientation and authorization are separate requirements.

Your route through the book

Unit	What you will produce	Suggested classroom time
1 · Industry evidence and decisions	A checked industry statement, document record and bounded environmental comparison	90 minutes
2 · How roads changed	A sourced timeline and explanation of materials, methods and purposes	75 minutes
3 · From materials to paving	A labelled flow map and a question about a product or process	90 minutes
4 · Working in a paving team	A role handover, responsibility decision and learning plan	90 minutes
5 · Read road documents and quantities	A checked plan/profile/section record and revised quantity calculation	120 minutes
6 · Investigate a paving trend	An evidence comparison and limited recommendation	90 minutes

These proposed learning times include supported Skills for Success practice. They are not a provider timetable or technical-training hour requirement. An instructor may divide a unit into shorter sessions and record additional support time.

Bring all nine skills into useful work

Reading helps you locate conditions and compare versions. Writing leaves a record another person can use. Numeracy makes quantities and comparisons checkable. Digital helps retrieve, organize and revise information. Problem Solving helps define an issue before choosing a response. Communication checks shared understanding. Collaboration brings contributions together. Adaptability helps revise a plan when conditions change. Creativity and Innovation helps develop and test a useful improvement. Each task names the skills most visible in that activity; every learner needs an individual opportunity to demonstrate them.

The activities use identifiers such as LA3-02. Keep the identifier with your work, the date and the support you used. You may begin with paper, a spoken explanation, a diagram or an accessible tool where the task allows. When the outcome is writing, keep a text whose wording you control. When the outcome is digital tool use, demonstrate the action on an actual device; a paper plan can prepare you for it but does not show the action itself.

A routine for every unit

Read the task and name the required decision. Mark the facts, units, dates and limits. Make a first attempt before consulting answer guidance. Compare your reasoning with a partner or instructor, then revise a specific feature. Keep both the original and the revision when possible. A changed answer is useful evidence when you can explain what changed and why.

The local feedback descriptions are **Not observed**, **Not yet demonstrated**, **Demonstrated with support** and **Demonstrated independently**. They describe this task under these conditions. They do not assign an official trade qualification or a permanent level to a person. A future workplace application is a transfer plan until someone observes the actual use in context.

Learning together in B.C.

Learners bring different work, family, community and language experiences. Use that knowledge if you choose, and keep personal information private when you prefer. Nobody is expected to speak for a Nation or community. A project affecting a particular place needs the appropriate relationships, permissions and local knowledge; a classroom map cannot supply them. Ask before recording, sharing or displaying another person's contribution.

All practice projects, dimensions, schedules, supplier messages and cost models are labelled fictional. Public facts have dates and a research record in the back matter. When a document lacks the information needed for a decision, name the missing information and the person or authority who should confirm it. That is part of competent work.

1 · Industry evidence and decisions

Read a number with its label

A statement about an industry needs a measure, geography and period. A count of people employed is different from a count of employers. A forecast of openings over ten years is different from the number employed in one year. Construction includes building work and civil engineering as well as paving. A broad construction figure provides context; it cannot be renamed an asphalt-paving total.

Evidence card I1 · B.C. construction. WorkBC's Construction industry profile, retrieved 29 October 2026, reports the following. The 2024 values are attributed on the page to the Labour Force Survey. The openings forecast is attributed to the B.C. Labour Market Outlook.

Measure	Period	Reported value	Population or scope
Employment	2024	246,750	B.C. construction industry
Full-time share	2024	92%	People employed in B.C. construction
Part-time share	2024	8%	People employed in B.C. construction
Full-time share	2024	80%	People employed across B.C. industries
Part-time share	2024	20%	People employed across B.C. industries
Forecast job openings	2025–2035	87,280	B.C. construction industry

The public page does not supply an asphalt-specific employer count in this extract. Shares are published rounded values; multiplying them by an employment total produces an approximation, not a newly observed headcount. Forecasts describe expected demand under assumptions and do not guarantee a job or a person's hours.

Evidence card I1b · Road-industry business locations. Statistics Canada's table 33-10-1095-01, *Canadian Business Counts, with employees, December 2025*, released 13 February 2026, reports **427 active statistical locations with employees** in B.C. under **NAICS 237310, Highway, street and bridge construction**. This is a closer road-industry classification than all construction, but it includes more than asphalt paving. A company with several statistical locations can be counted several times. Locations are not a count of unique employers, people or projects.

Employment-size category	Reported B.C. statistical locations
1–4 employees	175
5–9 employees	63
10–19 employees	72
20–49 employees	62
50–99 employees	32
100–199 employees	18

Employment-size category	Reported B.C. statistical locations
200–499 employees	4
500 or more employees	1
Total with employees	427

The table's notes explicitly say not to use the size ranges to calculate total employment, and advise against using these counts as a time series because methodological changes can affect differences. The retained machine record has reference-period code `2025-07`; the publisher labels the semi-annual table **December 2025**. Keep both in a source record rather than silently rewriting the downloaded field.

Worked example. The difference between the construction full-time share and the all-industry share is `92% − 80% = 12 percentage points`. It is not a 12% increase in employment. An approximate construction part-time count would be `246,750 × 0.08 = 19,740 people`, using a rounded share. A careful sentence keeps the word *approximately* and the 2024 construction scope.

LA1-01 · Report what the data can support

Write a four-sentence industry note using I1: one employment statement; the full-time-share comparison; one forecast statement; and one limitation. Correct this claim: “There are 246,750 asphalt employers in B.C. and 87,280 guaranteed new paving jobs.” Then use I1b to check the sum of the eight size categories, find the difference between the 1–4 and 5–9 categories, and find the 1–4 category's share of all 427 locations to two decimal places. Explain why neither adding category midpoints nor subtracting 427 from 246,750 gives a valid workforce result. Mark the publisher, period, measure and scope beside each number. **Skills:** Reading, Numeracy, Writing.

LA1-02 · Retrieve and preserve a source record

Using the public WorkBC link in the research record, locate one I1 value on a device. Record the page title, publisher, address, access date, measure and reporting year. Compare it with I1; a changed page does not make the retained dated extract disappear. Save and reopen your note. If access is unavailable, use the printed card for reading and schedule a later digital turn. **Skills:** Digital, Reading, Adaptability.

Separate law, contract information and funding

Evidence card I2 · Workplace orientation. B.C.'s Occupational Health and Safety Regulation, sections 3.22–3.25, defines a new worker to include someone new to a workplace and people affected by changed workplace hazards. Section 3.23 requires the employer to provide workplace-specific health and safety orientation and training before a young or new worker begins work. Topics include supervisor contact information, rights and responsibilities, hazards, emergency procedures, and instruction and demonstration of the work task. Section 3.24 addresses additional training when observation shows it is needed or when requested; section 3.25 requires records. This card identifies selected provisions; it is not the complete orientation.

Evidence card I3 · Highway specifications. B.C.'s Standard Specifications for Highway Construction page, updated 1 April 2026, describes the specifications as a standard reference in construction contracts. It identifies the 2025 edition as adopted 1 February 2025 and an amendments appendix updated 1 April 2026.

The page says modifications for local conditions are specified in special provisions for each contract. A current public edition is a starting point: the signed contract, drawings, amendments and special provisions establish which requirements apply to a particular job. A specification reference is not automatically a statute.

Evidence card I3b · A defined abbreviation. The 2006 *Design Build Standard Specifications for Highway Construction*, clause **502.03.01**, defines **EPS as End Product Specification**. In that clause, the contractor is responsible for construction-process quality control and the Ministry performs specified end-product audit sampling and testing for acceptance/rejection and payment adjustment. The definition is confirmed in PDF page 297, printed 1 of 22. That historical definition does not establish which edition governs a current contract, or prove the intended meaning of every unexplained use of EPS. In a geofoam discussion, EPS means expanded polystyrene; read the surrounding subject and definition.

Evidence card I4 · A completed project. The Kicking Horse Canyon Phase 4 project pages report completion on 31 July 2024. The project overview lists Government of Canada investment of **$215.19 million** and B.C. government investment of **$385.58 million**. The procurement page records the Province awarding the design-build contract to **Kicking Horse Canyon Constructors** on 6 November 2020. That team consisted of Aecon Group Inc., Parsons Inc. and Emil Anderson Construction. Funding is not the same as a contractor's profit or an individual worker's pay. The public summary is not the project's complete contract.

LA1-03 · Match a question to a document

For each question, choose I2, I3, I3b, I4 or “additional document needed,” and explain the limit: What training does a new worker need before starting? Which specification edition governs a particular signed contract? Who are the two reported public funders? What is the contractor's profit? What does EPS mean in clause 502.03.01 of the 2006 specifications? What further context is needed to interpret EPS in an old note that gives no definition? Create a document register with title, issuer, date/version, status and question still open. **Skills:** Reading, Problem Solving, Writing.

LA1-04 · Check the funding statement

Add the two I4 contributions and calculate the federal share of that sum to two decimal places. Explain why “more than $600 million” is consistent with the listed values. Rehearse a 45-second explanation distinguishing funder, contracting authority and contractor. Your partner checks the roles and units, then you swap. **Skills:** Numeracy, Communication, Collaboration.

Compare environmental claims within a boundary

Asphalt and concrete are pavement materials, not complete environmental outcomes. Compare a defined service, time period and system boundary. A material-production figure alone leaves transport, construction, maintenance, use and end-of-life outside the answer. A reusable material is not automatically the option with the lowest total impact. The amount recovered, replacement of new material, processing energy, transport and required performance all matter.

Evidence card I5 · Material recovery, a bounded comparison. The U.S. EPA's March 2004 *Hot Mix Asphalt Plants* chapter describes asphalt mixtures that may include reclaimed asphalt pavement, or RAP, alongside aggregate and asphalt cement. FHWA's undated historical *Recycled Concrete Aggregate* review describes concrete being recycled as aggregate, principally for base material in the review's U.S. context. These are two evidenced recovery pathways. Neither card provides a matched B.C. life-cycle emissions comparison, equal

pavement designs, haul distances or a common service life. They support “both materials can enter recovery pathways,” not “either material is always greener.” The references describe mechanisms rather than current local market shares.

Practice decision brief, fictional. A community facility is considering two pavement options for the same 50 m by 6 m access route over a proposed 20-year comparison period. Both must meet the same approved function and loading requirements, which a qualified designer would establish. Option A proposes asphalt with some RAP; Option B proposes concrete with some recycled aggregate. The proposer supplies no verified quantities, maintenance plan, emissions factors or transport distances. The group must decide whether a claim of “30% lower whole-life emissions” has been established. No numerical environmental advantage is supplied as fact.

LA1-05 · Make the strongest supported comparison

Create a comparison grid with material production, transport, construction, maintenance/use and end-of-life rows. For each option, distinguish the recovery pathway supported by I5 from information missing in the fictional brief. Write a conclusion of 60–90 words. Include the common function, 20-year proposed period, one supported similarity and at least three data needs. **Skills:** Reading, Problem Solving, Writing.

LA1-06 · Revise a claim when the scope changes

A proposer now says the “30%” applies only to factory production per tonne and compares different product quantities. Explain why that does not answer the whole-route, 20-year question. In pairs, redesign the comparison request so the next submission could be evaluated. Each learner records one contribution and one revised sentence. **Skills:** Numeracy, Collaboration, Adaptability, Creativity and Innovation.

Keep your industry note, source record, document register, funding calculation and comparison request. An instructor checks the evidence and the questions you leave open, not whether you force a confident answer from incomplete information.

2 · How roads changed

Ask what a road was built to do

Road history is a history of materials, movement, power and relationships with place. A route can serve travel, trade, access to resources, settlement or military movement. Its purpose can change. A road that enables one activity may also affect land, water, access and existing travel routes. A timeline should identify whose account is being used and avoid treating one engineering tradition as the beginning of all movement.

In B.C., a classroom discussion of road history does not give permission to map or publish community-held routes or knowledge. Use the public evidence cards below. Invite local knowledge only through the appropriate relationships and agreed purpose. The lesson asks you to explain change, not to speak for a community.

Evidence cards for a timeline

H1 · Via Appia. UNESCO describes the Appian Way as a major Roman road constructed and developed from **312 BCE to the fourth century CE**. Its initial strategic purpose included military conquest. It later supported connected settlements, agriculture and trade. Engineering works included the road and associated structures. One road can have several purposes over time; “Roman” does not mean “the world's first road.”

H2 · Alaska Highway. Public Services and Procurement Canada's history, updated **16 December 2025**, explains that the U.S. Army Corps of Engineers built the road during the Second World War to provide land access to Alaska. It identifies local First Nations' vital role as guides. The pioneer road was completed in **1942**, and the permanent location in **1943**. Responsibility for the Canadian portion transferred to National Defence in **1946**. Official civilian non-military use began in **1948**. Further responsibility changes occurred in **1961**, **1964** and **1992**. These are different events, not competing dates for the same event.

H3 · Post-war highways. FHWA identifies the U.S. Federal-Aid Highway Act of **1956** as a major step in the Interstate system. Its historical discussion connects the system with national planning and everyday movement of goods and people. The 1950s were within the Cold War period. A date and a national-security context do not prove that every new road had one motive; use evidence for the particular project.

H4 · Present resource roads. B.C.'s resource-road information describes a network used for resource industries and access. A resource road is not necessarily paved, and a logging road is not necessarily a corduroy road. Its owner, administrator, condition and access rules need to be established from the relevant record. A present-day category does not describe every historical method used to build it.

LA2-01 · Build a timeline that preserves meaning

Place 312 BCE, 1942, 1943, 1946, 1948 and 1956 in chronological order. Label each event and its evidence card. Identify two different meanings of “completed” in H2. Write one sentence explaining why a timeline alone cannot show every social or environmental effect of roadbuilding. **Skills:** Reading, Writing, Problem Solving.

Materials and methods solve different problems

An earth route uses the ground surface, sometimes shaped or improved. Stone and gravel add mineral material. Cut or shaped stone can form a durable surface where the design, foundation and maintenance suit the use. Wood has also been used: a **corduroy road** is a historical construction form using closely laid logs across the travel direction, often associated with soft or wet ground. The log pattern explains the name. It is a method, while *logging road* describes a road's use; the categories overlap in some histories but are not synonyms.

Asphalt pavement combines aggregate with a bituminous binder. Concrete pavement uses a cement-based binding system. These materials have different production and construction processes. Do not confuse asphalt cement—the binder in an asphalt mixture—with Portland cement used in concrete. A road may have several layers and may be repaired with a different material from its original surface. Appearance alone is not enough to identify its entire construction.

Modern asphalt placement commonly involves delivery of mixture, controlled placement through a paver and screed, and compaction using rollers, with associated preparation, checks and finishing. Mechanization changes how material is moved and shaped; it does not remove the need for a suitable foundation, drainage, specifications or skilled people. This unit describes the change in method. Operating settings belong to the relevant supervised training and approved documents.

Worked example · Correct a category mistake. A museum draft says, “All logging roads were corduroy roads, and modern roads replaced wood with asphalt.” The first clause confuses purpose with method. The second implies one universal sequence. A better label is: “Corduroy construction used logs across a route. It was one historical way of addressing difficult ground. Logging roads have used varied surfaces and construction methods; modern asphalt paving is one distinct technology.”

LA2-02 · Match material, method and purpose

Create three columns headed material, method and purpose. Sort these terms: earth; cut stone; gravel; timber logs; concrete; moving harvested timber; military access; asphalt mixture; closely laid crosswise logs; delivery/placement/compaction; travel to a community facility. Explain the difference between asphalt cement and Portland cement. Explain why a purpose can be served by more than one material. Add one question you would ask before classifying an unfamiliar road from a photograph. **Skills:** Reading, Communication.

LA2-03 · Explain a road without inventing its history

Fictional archive label: “Photo R17, date not recorded. A line of logs crosses a wet-looking route. A truck appears in the background. No owner, purpose or place is recorded.” Write a 50–80-word description separating observation, reasonable interpretation and unknown information. A partner reads it and identifies any unsupported claim. Revise one sentence. **Skills:** Writing, Collaboration, Adaptability.

Explain why demand changed

War can accelerate road construction, as H2 shows. Post-war planning and Cold War concerns formed part of the setting for major highway development, while trade, industry, settlement and daily access also influenced demand. The same road can support freight, recreation and local travel. “More roads” is not a complete measure of benefit: maintenance, alternatives, land use and the people affected belong in the question.

LA2-04 · Compare two explanations

Compare these claims: “The Alaska Highway began as a wartime access project” and “All highways were built only for war.” Use H2 and H3 to support, limit or reject each claim. Calculate the interval from the 1942 pioneer-road completion to the 1948 opening to civilian non-military use. Explain why that interval does not mean no work occurred between those dates. **Skills:** Reading, Numeracy, Problem Solving.

LA2-05 · Connect a historical method to a modern sequence

Make a two-panel classroom display: one panel explains a corduroy route using the supplied definition; the other explains modern asphalt delivery, placement and compaction. Label material, method, purpose and one unknown in each. Use words, a simple sketch or an accessible digital layout. Do not provide a construction recipe. Each partner explains one panel and checks the other's labels. **Skills:** Creativity and Innovation, Collaboration, Communication; Digital if a tool is actually used.

LA2-06 · Independent transfer: a careful public caption

Write a 90–120-word caption titled “Roads change as needs and methods change.” Include a dated example, a material/method distinction, a reason for road growth and a boundary on what the evidence establishes. Add a source note from the research register. Keep the first and revised versions. Your next step may be a local public-archive enquiry, but record it as a proposed enquiry until evidence is actually obtained.

3 · From materials to paving

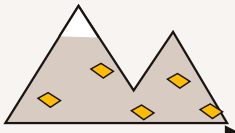
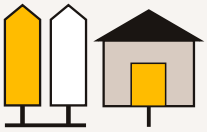
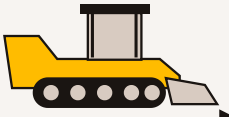
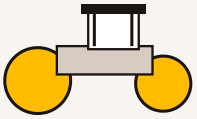
Follow the material through the system

Paving technology connects material sources, manufacturing, transport and placement. Begin with the function of each stage. A pit or quarry supplies mineral material; processing prepares aggregate to specified characteristics; a plant combines selected materials; haul vehicles move the mixture; paving equipment places and shapes it; rollers compact the placed layer. Preparation, testing, traffic arrangements and finishing also matter. A flow map is a way to understand relationships, not permission to operate equipment.

ROADBUILDING / MATERIALS THROUGH THE SYSTEM

Follow the material, step by step.

A plant route, a return route, and a distinct route on existing pavement.

01	02	03	04	05
PREPARE	**MIX**	**HAUL**	**PLACE**	**COMPACT**
				
Pit / quarry	**Batch OR drum plant**	**Haul truck**	**Paver + screed**	**Rollers**
Processed aggregate	Aggregate + binder	Move the mixture	Place the mixture	Compact placed material

PLANT RETURN ROUTE

Removed pavement → **RAP**
Reclaimed asphalt pavement → **Back to a plant**
where used

DISTINCT ROUTE / HOT-IN-PLACE



A train works on existing pavement.

Keep this route distinct from taking reclaimed material back to a plant.

Know what the ingredients do

Aggregate is mineral material such as processed sand, gravel or crushed stone. A **pit** generally obtains unconsolidated deposits; a **quarry** extracts rock that may need crushing and screening. Material availability does not establish suitability: the required properties come from the approved design and specification. A short haul distance alone cannot answer whether a source meets the project requirements.

Asphalt binder, also called asphalt cement in many documents, binds the aggregate in asphalt mixtures. It is distinct from the Portland cement used in concrete. **Additives** and modified binders can change mixture characteristics for a defined purpose, but their selection and proportions require the applicable approved information. Fuel used to heat a production process is not automatically an ingredient in the pavement mixture. A delivery label must identify the actual product rather than relying on the loose word *oil*.

Technology evidence card T1. EPA's *Hot Mix Asphalt Plants* description identifies sized aggregate, liquid asphalt cement and potentially reclaimed asphalt pavement as components of hot-mix asphalt. It describes batch and drum-mix production. Its technical account is used here for the process distinction; its historical U.S. production totals and operating details are not current B.C. specifications.

LA3-01 · Explain the ingredients to a new learner

Write a three-sentence explanation that distinguishes aggregate, asphalt binder and fuel. Add a sentence explaining the conditional purpose of an additive and why its use needs approved information. Distinguish a pit from a quarry, then name one question about an aggregate source that proximity alone cannot answer. Explain why “cement” needs context when comparing asphalt and concrete. **Skills:** Reading, Writing, Communication.

Distinguish batch and drum production

A **batch plant** prepares identifiable batches: aggregate is proportioned and combined with binder in a mixer. A **drum-mix plant** uses a continuous process in which material flows through the plant. Drum designs differ; do not treat every drum layout as identical. The beginner distinction is batch quantities versus a continuing material stream, not “one uses a drum and the other has no drum.” A batch plant can have a drying drum before its mixer.

Fictional classroom production records. Plant B's teaching log lists batch 01, batch 02 and batch 03, each with a separate record of ingredients and release. Plant D's teaching log records a continuous feed period and time-stamped checks. These are deliberately simplified descriptions. Neither log provides an approved mix design, production rate or operating instruction.

Worked example. “Plant B has a drying drum, so it must be a drum-mix plant” is an invalid inference. Ask where and how the mixture is produced. B's separate batch records support a batch process in this case. The physical presence of one part does not classify the whole system.

LA3-02 · Compare two process records

Identify the production type indicated by each fictional log. Draw or describe the common stages and the key difference. Add a correction to the worked-example claim in your own words. In pairs, one learner explains the map while the other checks whether it accidentally implies that a batch plant lacks a drying drum. Swap roles. **Skills:** Reading, Problem Solving, Collaboration.

Treat product codes as information to verify

A product code can identify a grade system or a product, but it does not supply the whole application instruction. A penetration-grade designation and a performance-grade designation are different ways of describing binder characteristics. The numbers are not interchangeable merely because both appear beside asphalt. The purchaser and operator need the approved product, governing specification, supplier record and relevant safety information.

Fictional document-check case. An archive card contains the labels “85–100 Pen” and “PG 72–22,” and a handwritten note says “altered oil.” A new practice request says only “send the usual oil.” No current approved binder, project specification or supplier product sheet has been attached. These labels are for recognition and clarification; the case does not recommend either grade for B.C. work. Do not infer a mixing temperature, quantity or handling procedure from them.

LA3-03 · Write the clarification before the order

Make a four-item information request that would identify the intended product and its governing documents. Explain why “usual oil” and an archive grade label are insufficient. Read your request to a partner, who checks whether you have accidentally approved a product. Keep it as a classroom draft. **Skills:** Writing, Communication, Problem Solving.

Separate the recycling routes

Reclaimed asphalt pavement (RAP) is asphalt pavement material recovered for possible reuse. A plant-recycling route takes recovered material through suitable processing and an approved production process. Reuse is conditional on the design and quality requirements. A claim that all recovered material can be added without testing is unsupported.

Hot-in-place recycling treats an existing asphalt surface on the road using a coordinated equipment train. Its broad purpose is to reuse existing material in place through a specialist process. It is different from hauling recovered material to a plant. A milling machine's removal of material is also not proof that the removed material has been successfully recycled. Track the actual route and evidence for reuse.

An **asphalt train** is a coordinated sequence of equipment and roles. In ordinary placement, haul, paver/screed and compaction functions connect. A specialist hot-in-place train has its own process and equipment. The word *train* does not mean every paving job uses the same machinery, controls or sequence.

LA3-04 · Trace two reuse pathways

Draw two labelled routes: recovered material sent to a plant, and a specialist hot-in-place process on the existing road. Mark where the existing material is recovered or reused. Add one document or check needed before calling either route approved. Explain why “milled” and “recycled into a new pavement” are different claims. **Skills:** Reading, Creativity and Innovation, Problem Solving.

LA3-05 · Calculate a fictional material record

A classroom demonstration model uses **80 counters for new aggregate, 15 for recovered mixture and 5 for binder**, for 100 counters in total. Counters are a visual model, not a mix recipe; the recovered mixture already contains its own materials. Find each category's share of the model and check the sum. Explain why adding the five binder counters to the 15 recovered-mixture counters does **not** establish the real binder content. **Skills:** Numeracy, Reading.

LA3-06 · Respond to a changed process description

A team initially labels a photo “RAP going to a plant.” The caption is corrected to “specialist hot-in-place train.” Revise the material-flow card and give a handover identifying what changed, what remains accurate and what further process information would be needed. Each learner makes one recorded revision. **Skills:** Adaptability, Collaboration, Communication.

LA3-07 · Make a retrievable technology note

Create a one-page document with the two plant distinctions, ingredient roles, two recycling routes and the unresolved product request. Save, close and reopen it; check that another learner can find its title and version. Offer a paper route for composing, while keeping actual Digital evidence separate. Finish by explaining one function without providing an operating setting.

4 · Working in a paving team

Understand the role and the handover

A paving team coordinates people whose tasks depend on one another. The driver delivers material; the paver operator manages the machine's placement function; the screed role attends to the screed's contribution to the placed mat; rakers support hand placement and finishing; roller operators perform the assigned compaction work; milling operators work with equipment that removes or profiles existing material. Job titles and boundaries vary. A job title alone does not establish a person's training, authorization or responsibility on a particular site.

Fictional team brief W1. At the Willow Reach training centre, a classroom team is preparing an explanation of a paving job. Jo coordinates the display. Ren checks the delivery/placement sequence. Ash prepares the screed-and-raker explanation. Kim checks the compaction and milling distinctions. The team has public introductory descriptions, but no site plan, machine manuals or permission to enter a worksite. The display must explain broad roles and show where one role needs information from another.

LA4-01 · Make a role and dependency map

Map haul delivery, paver, screed, raker, roller and milling roles. For each, write one broad function and one information handover. Identify two assumptions W1 does not authorize. Each learner explains a different role before the group combines the map. **Skills:** Reading, Communication, Collaboration.

Treat health and safety as a shared system with defined duties

Health and safety includes planning, suitable equipment and systems, instruction, supervision, reporting and review. It is not satisfied by telling a worker to “be careful.” A new person needs to know the supervisor, relevant hazards, emergency arrangements and how the actual task is done safely. Employers have responsibilities that cannot be replaced by a worker's confidence or a classroom certificate.

B.C.'s orientation provisions require workplace-specific training before a young or new worker begins work. A new worker can be an experienced person who is new to that workplace or affected by changed hazards. Additional orientation and training must be provided when observation shows a need or when the young or new worker requests it. Records are required. The complete workplace program and applicable legislation must guide actual work.

Fictional case W2. A learner arrives for a first workplace visit after completing classroom vocabulary. A host says, “You know the equipment names, so join that crew.” No supervisor contact, task demonstration or emergency information has been provided. Another host offers to stop the assignment and arrange the workplace orientation. The learner wants to participate but does not know the site's communication arrangements.

LA4-02 · Choose the responsible next step

Explain why the second host's approach addresses a real gap. Draft a brief request for orientation and task clarification. Identify information the employer must provide and a question the learner can ask. Do not improvise signals or a route through live equipment. **Skills:** Problem Solving, Writing, Communication.

Check licences and authorization precisely

A driver's licence, admission to a training course, equipment competence and site authorization answer different questions. ICBC identifies Class 5 as the common passenger-vehicle licence. A provider may publish its own admission requirement. Neither statement proves that the holder may operate every paving machine or drive every vehicle combination. Determine the actual vehicle, activity and location, then verify the applicable licence class, restrictions, endorsements, training and employer authorization through the responsible sources.

Document-check case W3, using a real provider statement and fictional decision. IUOE Local 115's public Asphalt Paving Laydown Technician page, retrieved 29 October 2026, lists a valid Class 5 or 7 driver's licence among its admission prerequisites. A fictional learner reads this and says, “Then Class 7 authorizes me to drive any haul truck at work.” The page is a course-admission source; it is not a legal ruling on that vehicle or a workplace assignment.

LA4-03 · Separate four kinds of permission

Create a table for course admission, driver's licence, equipment competence and site assignment. State the question each answers and who or what record can confirm it. Correct the learner's claim without replacing it with another unsupported universal rule. **Skills:** Reading, Problem Solving, Writing.

Practise professionalism through observable actions

Professionalism includes preparing for the agreed work, communicating changes, checking information, respecting people and taking care with records and materials. Pride in a project can be shown through a correction made before a mistake is passed on. It does not require pretending to know something or remaining silent when instructions are unclear.

Worked handover. “The map is finished” leaves the receiver guessing. A useful handover is: “The haul, paver and roller functions match our classroom brief. The licensing question is unresolved. The source note is saved as version 2. Please check the role labels; do not describe the draft as a site procedure.” The receiver restates the completed and pending parts before continuing.

LA4-04 · Repair a rushed handover

Jo says, “Everything is approved—just print it,” although W3 is still unresolved. Give a respectful correction that protects the deadline and the accuracy of the work. Agree a revised print note and record each person's contribution. **Skills:** Communication, Collaboration, Adaptability.

Plan development without promising job security

A worker may develop from entry tasks toward equipment roles, specialist functions, supervision or other pathways through training, experience and appropriate assessment. Progress is not automatic or identical for everyone. Ask what evidence, supervision and opportunities the actual employer or provider requires. A forecast of industry demand cannot guarantee a particular person's employment. Season, project timing, skills, location, employer needs and economic conditions can affect work.

Paving work may involve outdoor conditions, dust and fumes, noise, night or weekend schedules and seasonal changes. A job enquiry should ask about the actual schedule, hazards, controls, travel, orientation and employment terms. Do not assume a fictional schedule establishes lawful hours, overtime or pay rules.

Fictional schedule W4. A classroom planning record lists a Monday period from 18:00 to 02:00, including a 30-minute unpaid break for arithmetic practice. Tuesday is cancelled because of a changed project plan. Wednesday has six paid hours. These are invented conditions, not a recommended work arrangement or statement of employment law.

LA4-05 · Check time and revise the plan

Find Monday's elapsed and paid time, then the total paid hours recorded for Monday and Wednesday. Write two questions about a real employer's terms that this classroom record cannot answer. Revise a weekly learning plan so a cancelled day does not become falsely recorded practice. **Skills:** Numeracy, Adaptability, Writing.

Understand organized labour and employment documents

A union can represent workers in collective bargaining with an employer. A collective agreement records negotiated terms for the bargaining relationship; its application depends on the employer, bargaining unit, dates and the actual agreement. B.C.'s Labour Relations Board maintains a searchable directory of filed agreements. An old agreement found online is not automatically the current agreement for a new job. A training provider's fee page is also different from an employment agreement.

LA4-06 · Prepare a useful enquiry and growth plan

Create an enquiry with four questions: applicable employment/collective-agreement information; expected working conditions; training and authorization for a desired role; and how progression is assessed. Add one sentence explaining collective bargaining and one explaining why industry demand or course attendance does not guarantee employment. Add a two-step learning goal with an observable result and a check date. Use fictional names; do not send the enquiry. On a device, save and reopen it with a clear version name. A partner suggests one improvement and you decide whether to use it. **Skills:** Digital, Writing, Creativity and Innovation, Communication.

5 · Read road documents and quantities

Identify the project before the specification

A project may be described by use—commercial, industrial or residential—or by jurisdiction and delivery, such as a provincial highway project. These categories answer different questions and can overlap. A private industrial site is different from a provincially administered highway, even if both use asphalt. Owners can include provincial, federal, municipal or private organizations. A funder, designer, contractor and owner may be different parties.

Fictional practice project R1 · Cedar Learning Access, drawing P-01 revision B. The fictional Cedar Learning Society owns a short access route on its own practice property. Valley Surface Ltd. is the fictional contractor. The classroom project brief classifies the work as a private facility access project. A fictional community fund contributes money but does not become the owner. There is no real construction instruction or permission attached to this case.

The practice document register contains P-01 revision B, a plan/profile/section sheet; Q-01 revision B, a quantity record; and SP-01 revision B, a teaching specification card. An earlier P-01 revision A is retained for comparison and marked superseded. In a real project, the contract and authorized document system determine applicable requirements; a familiar document title does not establish that it governs every road.

Document families include B.C.'s provincial highway specifications, **Master Municipal Construction Documents (MMCD)**, a local government's project requirements, and building-specific architectural or civil drawings and specifications. MMCD's publisher produces standardized municipal construction documents and updates. A municipal or private project may adopt particular documents through its contract; naming MMCD does not establish that every edition or clause applies. Provincial specifications likewise need the applicable contract version and special provisions. Record the issuer, edition, project adoption and any authorized change before relying on a requirement.

LA5-01 · Sort use, ownership and authority

Identify the project use, owner, contractor and funder in R1. Explain why “private” and “asphalt” do not tell you which public standard or special provision applies. Compare provincial, MMCD, local-government and building-specific document families: what project-adoption evidence would you check before using any of them? Name the record you would need before treating a requirement as authorized. **Skills:** Reading, Problem Solving, Writing.

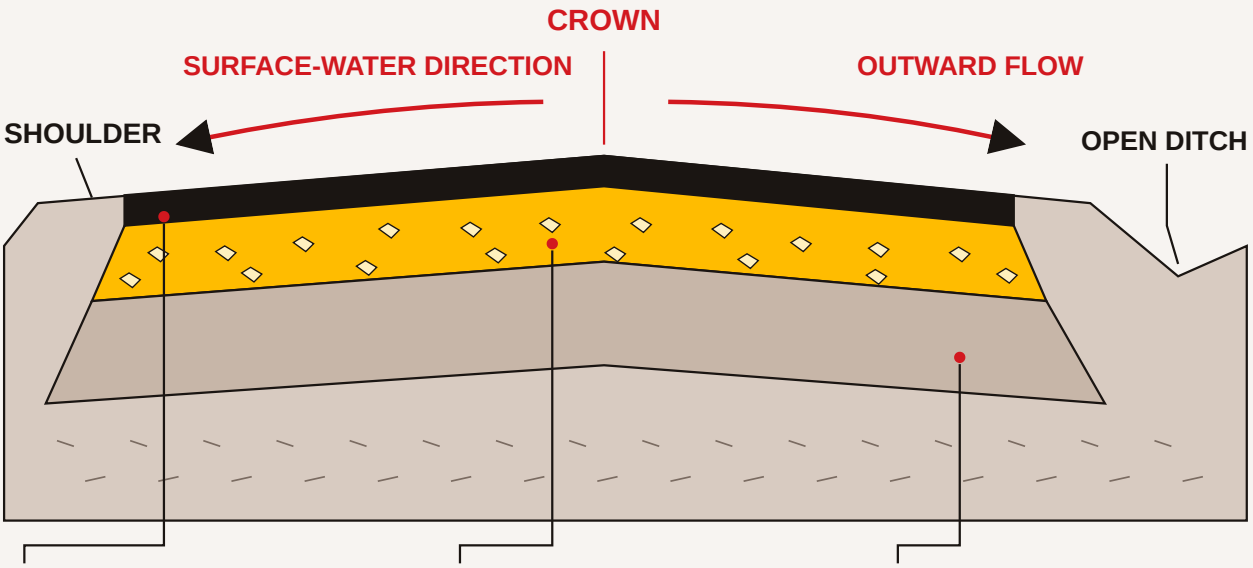
Read three views of the same place

A **plan** looks down on a project and shows its horizontal arrangement. A **profile** shows elevations along a chosen line, commonly using chainage or stationing to locate points. A **cross-section** cuts across the route and shows its transverse form and layers. A number must be associated with the correct view, direction and unit. Plan width, profile rise and layer thickness are different quantities.

ROADBUILDING / READ A CROSS-SECTION

See how the layers work together.

A conceptual cut across a road with an open-ditch edge.



CONCEPTUAL TRAINING VIEW

Depths exaggerated • No design dimensions • Curb-and-gutter edge not shown

A **crown** is a raised transverse form that can direct water toward the sides. **Superelevation** describes a cross-slope arrangement used on curves; its design depends on the project. **Runoff** is water moving from the surface. An open ditch and shoulder arrangement differs from a contained curb-and-gutter edge. A sidewalk serves a different function from the vehicle surface. A grader is equipment used in earthwork or grading, not a layer in the finished section. Drainage and preparation must be coordinated before paving; a drawing needs more than a surface outline to establish a buildable design.

P-01 revision B · Complete fictional drawing data. The rectangle extends from chainage 0 m to 50 m. Its constant width is 6 m. The new uniform layer is 40 mm thick for this calculation exercise. No widening, taper, overlap, waste allowance or material loss is included. The exercise asks for geometric volume only. Use the written dimensions; a screen image is not a measuring scale.

Profile point	Chainage along centreline	Illustrative elevation
Start	0 m	100.00 m
Midpoint	25 m	100.25 m
End	50 m	100.50 m

The section names surface course, aggregate base and prepared subgrade. Their presence in a conceptual sketch does not specify material properties, layer thicknesses other than the given calculation layer, drainage design or acceptance criteria. All elevations and dimensions here are invented. The profile arithmetic is not a recommendation for road grade or crossfall.

LA5-02 · Choose the right view

For route length, width, centreline rise and layer thickness, identify the relevant view and unit. Explain why the profile elevation 100.50 m is not the layer thickness. Sketch three boxes labelled plan, profile and cross-section and put the matching information in each. **Skills:** Reading, Numeracy, Creativity and Innovation.

LA5-03 · Describe the road form

Using the visual and paragraph, explain crown, shoulder, open ditch, curb/gutter, sidewalk, runoff and superelevation. Put “grader” in the correct category. Choose one drainage question and one preparation question that the conceptual drawing leaves unanswered. **Skills:** Reading, Communication, Problem Solving.

Calculate a quantity you can check

Worked example, P-01 revision B. Convert thickness first: `40 mm ÷ 1,000 = 0.040 m`. Plan area is `50 m × 6 m = 300 m²`. Geometric layer volume is `300 m² × 0.040 m = 12 m³`. Estimate: a 300 m² layer one-tenth of a metre thick would occupy 30 m³; a 0.04 m layer is less than half as thick, so 12 m³ is plausible. Units check as `m × m × m = m³`.

The profile rises `100.50 − 100.00 = 0.50 m` across 50 m of stated horizontal distance. For this drawing-reading exercise, rise/run is `0.50 ÷ 50 × 100 = 1%`. This centreline grade is not the transverse crown or an acceptance limit. A real quantity may need a different geometry and an approved measurement method; our rectangular model has stated simplifications.

Q-01 revision B · Fictional mass model and delivery records. For this exercise only, the instructor supplies a conversion factor of **2.00 tonnes per cubic metre**. It is an invented teaching factor, not an asphalt density specification. The planned model mass is `12 m³ × 2.00 t/m³ = 24.00 t`. Three classroom records list 8.00 t, 8.00 t and 7.60 t. These are fictional record inputs, not a real delivery or placement measurement.

LA5-04 · Check the quantity and record

Independently reproduce the area, thickness conversion, volume and model mass with units. Sum the three records and calculate their difference from the planned model mass. Explain why that numerical difference alone cannot establish placed thickness, compaction, payment entitlement or rejected work. **Skills:** Numeracy, Reading, Writing.

Use the applicable version and conditions

SP-01 revision B · Teaching specification card, fictional. Use the dimensions printed in the current classroom drawing. Record the revision with every calculation. Keep calculation quantities distinct from acceptance measurements. Refer conflicting instructions to the instructor acting as document coordinator before issuing a final classroom quantity note. This is the complete specification for the exercise; it does not adopt a real road specification.

Change notice R2, fictional. The coordinator issues P-01 revision C, extending the rectangle to 60 m while retaining 6 m width and 40 mm layer thickness. The endpoint illustrative elevation becomes 100.60 m; the same linear 1% profile is used for arithmetic practice. The mass model remains 2.00 t/m³. The three existing classroom records stay unchanged. Revision C replaces B for the new quantity calculation; retain B as the previous version.

LA5-05 · Recalculate after the revision

Find the new area, volume and model mass, and the increase from revision B. Reconcile the unchanged record total with the new model requirement. Explain why changing a filename without recalculating would leave misleading information. Make a versioned paper or digital quantity note with inputs, method, check, result and unresolved limits. **Skills:** Numeracy, Adaptability, Writing; Digital when performed.

LA5-06 · Review and hand over

Work in pairs: one checks geometry and units, the other checks document revision and claims. Swap responsibilities on the revised case. Give a handover that states the current revision, quantity, supplied factor, record total and the decision that remains outside this exercise. Each learner keeps a checked calculation and records one useful correction or confirmed detail. **Skills:** Collaboration, Communication, Problem Solving.

6 · Investigate a paving trend

Turn a promising idea into a testable question

A trend is useful to investigate when it responds to a defined need. A product described as *green*, *new* or *innovative* still needs evidence. Ask what problem it addresses, how it works, where it has been studied, what limits apply and who would approve its use. Something new to a local project may have been used elsewhere for decades. Publication date and location matter.

This unit compares material and design ideas without selecting a live road design. You will prepare a recommendation for further investigation using complete evidence cards and a fictional project brief. A careful recommendation can say that one option deserves a closer technical study while leaving design approval to the appropriate professional.

Compare surface and reuse approaches

Trend card N1 · Paving stones. Modular paving uses individual units rather than one continuous asphalt layer. Some systems are designed to manage water through joints and underlying layers; the word *paver* alone does not establish permeability. Ask about the whole pavement system, loading, drainage, accessibility and maintenance. A patterned appearance is not environmental evidence.

Trend card N2 · Porous asphalt. FHWA's April 2015 technical brief describes porous asphalt pavements with stone reservoirs as systems intended to provide a surface and manage stormwater. Water moves through interconnected surface voids into the underlying reservoir. This differs from an open-graded friction course on a dense pavement, where water moves through the surface course and out toward the side. The brief identifies potential clogging and the need for specialized maintenance, as well as site/design limitations. A material name does not guarantee infiltration at an unsuitable site.

Trend card N3 · Recycling. Recovered asphalt may enter an approved plant or specialist in-place process. Recovered concrete may be processed as aggregate. Potential reductions in new-material demand need to be examined alongside processing, transport, suitability, maintenance and end-of-life. “Recycled” is a property of the material pathway, not a complete environmental score.

LA6-01 · Improve three claims

Rewrite these statements so they are bounded and useful: “All paving stones absorb rain”; “Porous asphalt never needs drainage planning”; “Any recycled material is automatically the lowest-impact choice.” For each, identify a mechanism, a condition and one question for a qualified reviewer. **Skills:** Reading, Problem Solving, Writing.

Ask what a material changes

Trend card N4 · Epoxy, colour and modified asphalt. Epoxy systems, coloured surfaces and modified binders are families of products and applications. Colour can serve appearance or wayfinding, but it does not by itself establish structural performance, skid properties, durability or accessibility. A modified binder may be

selected for a particular performance requirement. The full product description, intended use, test evidence, compatible construction system and approved specification are needed before adoption. Do not turn a product family into one universal recipe.

Trend card N5 · Lightweight fill and "floating roads." Lightweight fill reduces the load imposed by an embankment compared with a heavier fill arrangement. Expanded polystyrene, or EPS, geofoam is one engineered lightweight-fill material. Its use beneath a road does not mean the road floats on open water. The soil, loading, stability, protection, groundwater and installation conditions require geotechnical design. A brand name or generic word such as *foam* is not a material specification. Pumice is another material encountered in lightweight-fill discussions, with properties and availability that must be checked for the actual source and use.

FHWA's **April 2017 Ground Modification Methods, Volume I** describes lightweight fills and EPS geofoam. It explicitly states that the publication is not a standard, specification or regulation. Its chapter explains that material properties vary and detailed design requires further testing and evaluation. The examples show that lightweight-fill ideas are not all recent inventions.

Evidence table N6 · One historical U.S. cost example. FHWA's 2017 manual reproduces a Milpitas, California project comparison credited to Cheng 2016. The total costs below include material, transport and longevity costs. They **exclude installation and contractor overhead and profit**. They are historical U.S. project figures, not current B.C. quotes. The table's dollar figures are retained in their source context; no currency conversion is supplied.

Material	Reported example total
Traditional soil fill	$563,000
Pumice rock	$633,000
EPS-block geofoam	$1,145,000
Expanded shale/clay	$490,000
Wood chips	$545,000
Tire-derived aggregate	$334,000

Worked example. In this table, pumice's reported total exceeds traditional soil fill by `$633,000 − $563,000 = $70,000`. That result answers a question about this historical comparison. It does not establish that pumice always costs $70,000 more, that the cheapest listed option meets a new site's design needs, or that the table includes every cost.

LA6-02 · Read the cost boundary

Calculate the difference between EPS-block geofoam and pumice in N6. Name three reasons the table cannot be used as a current B.C. quotation. Explain why price ranking alone cannot establish technical suitability. Keep units and the word *historical* in your conclusion. **Skills:** Numeracy, Reading, Problem Solving.

Separate an enhancement from a requirement

Trend card N7 · Patterned surfaces. Street-print or other patterned treatments can add a visual effect to a pavement. A visual enhancement may serve a project purpose, but performance claims still need evidence. Ask about the actual treatment, friction, wear, maintenance, access needs and compatibility with the

underlying system. A decorative pattern is not evidence of structural strengthening or permission to use a cultural design. Any community-specific image or knowledge requires appropriate permission.

LA6-03 · Make a product enquiry

Choose N4 or N7 and draft five questions for a fictional supplier. Include intended use, product identity, evidence, maintenance and approval/permission. Make one question specific enough that a vague marketing answer would be recognizable. Keep the enquiry local; no message is sent. **Skills:** Writing, Communication, Creativity and Innovation.

Apply the evidence to a defined case

Fictional project N8 · A learning-centre courtyard. A facility team is considering a 600 m² courtyard used by pedestrians and occasional authorized service vehicles. The team wants to improve surface-water management and readable wayfinding. No geotechnical investigation, infiltration test, approved loading design, maintenance budget or cultural-design permission has been supplied. The project budget is not yet set. A promoter offers porous asphalt, coloured asphalt or an EPS-supported pavement as three “green” options. These options address different questions and are not interchangeable solutions.

LA6-04 · Compare options against the actual need

Make a table for the three offers. State the problem each might address, evidence provided in N1–N7, missing project information and the reviewer needed. Select one or two ideas for further investigation, with reasons. Do not approve construction or claim a total environmental winner. Each learner contributes an evidence-based question before the group makes its shortlist. **Skills:** Problem Solving, Collaboration, Reading.

LA6-05 · Adapt to a new constraint

The fictional facility manager says there is no commitment to specialized pavement-cleaning equipment and asks for a maintenance plan before further consideration. Revise your shortlist or investigation plan. Explain why this new fact is relevant to porous asphalt and why it does not automatically approve another option. **Skills:** Adaptability, Communication, Problem Solving.

LA6-06 · Build an evidence brief and test its clarity

Create a one-page brief containing the project question, one proposed investigation, a supported benefit, a limitation, missing information and a next decision. Add source titles/dates from the research record and mark predictions as predictions. Save and reopen the file if using Digital. Ask a partner to identify what you are recommending and what you are leaving undecided; revise one unclear sentence. **Skills:** Writing, Digital, Communication, Creativity and Innovation.

Keep the claims you improved, historical comparison, supplier questions, option table and revised brief. A useful innovation record shows the question, evidence, trial or investigation needed and the limits of the conclusion.

Working reference and evidence record

Words used in this book

Term	Meaning in these lessons
Aggregate	Mineral material such as sand, gravel or crushed stone
Asphalt binder / asphalt cement	Binding material in an asphalt mixture; distinct from Portland cement
Batch plant	Production arrangement that makes identifiable batches
Drum-mix plant	Production arrangement using a continuous material/mixing process; layouts vary
RAP	Reclaimed asphalt pavement material considered for an approved reuse pathway
Hot-in-place recycling	Specialist process treating existing asphalt pavement on the road
Screed	Paver component that contributes to forming the placed mat
Compaction	Densification process performed according to the applicable approved procedure
Milling	Removal or profiling of existing material using suitable equipment
Chainage	Position measured along a defined project line
Plan / profile / cross-section	Top view / elevation along a line / cut across a route
Crown	Raised transverse road form that can direct surface water toward sides
Superelevation	Cross-slope arrangement associated with a curve; design is project-specific
Runoff	Water moving from a surface
Specification	Documented requirements; applicability depends on its legal/contractual context
Special provision	Project/contract-specific provision that may modify or add requirements
Revision	Identified version of a document
Functional comparison	Comparison based on an equivalent required service, period and conditions
Life-cycle boundary	Stages included and excluded in an environmental assessment
Percentage point	Difference between two percentages expressed on the same basis
EPS in the 2006 specifications	End Product Specification, explicitly defined in clause 502.03.01; identify the edition and context before applying the term
EPS geofoam	Expanded-polystyrene material for an engineered geotechnical application; an abbreviation's meaning still depends on context

Record evidence without overstating it

For each selected activity, record: activity ID; date; task and document version; what you did; support used; result; check; feedback; revision; and next practice. Mark the observed skill. A group product needs an individual contribution record. An accurate calculation needs units and assumptions. A digital claim needs actual tool use. A future application is a plan until it has been observed.

Activity	Evidence kept	Support and feedback	Next practice
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____

Sources and continuing questions

The research register provides public titles, issuers, dates, URLs, retained short excerpts and retrieval hashes. It distinguishes dated factual evidence from invented classroom inputs and records questions that need current local or technical confirmation. Use the register to check a claim, then keep the question and the evidence together. A link or a confident statement alone is not the complete answer.

Supplied technical document

British Columbia Ministry of Transportation, Construction and Maintenance Branch. *2006 Design Build Standard Specifications for Highway Construction*, adopted 20 February 2006. Section 502.03.01, PDF 297 / printed 1 of 22, defines End Product Specification (EPS). The supplied historical document is distinct from the current public specifications page below; its presence does not establish current contract applicability.

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The curated evidence register gives the teaching use and limits; exact selected excerpts and hashes support checking. The road-industry table extract preserves the nine source rows.